

Lightweight Speech Emotion Recognition Using Hand-Crafted Acoustic Features: Cross-Dataset Generalization and Gender-Bias Analysis

Group ID	F26-____	Track	B – Research & Development
Members	22L-7786 (Lead) 23L-3002	Instructor	Hajra Waheed

1. Problem Statement

Speech Emotion Recognition (SER) classifies a speaker's emotional state (e.g., angry, happy, sad, neutral) from voice alone, with uses in call-centre analytics, mental-health screening and voice assistants. Chowdhury et al. (2025) show that a lightweight ensemble (CNN + CNN-BiLSTM) trained on hand-crafted acoustic features reaches high accuracy while remaining far cheaper than large pretrained models. However, such models are typically evaluated by training and testing on the *same* dataset, which hides two practical risks:

- **Generalization:** performance may collapse on unseen speakers and recording conditions.
- **Bias:** accuracy may differ systematically between male and female speakers.

Hypothesis: a lightweight hand-crafted-feature SER model that performs well within a dataset will lose significant accuracy in a cross-dataset setting and show a measurable gender performance gap. This project tests that hypothesis and identifies which acoustic features matter most.

2. Dataset

Dataset	Type	Details	Role
RAVDESS (Kaggle)	Audio (.wav)	1,440 clips, 24 actors (12 M / 12 F), 8 emotions	Training, within-dataset testing, gender-bias check
TESS (Kaggle)	Audio (.wav)	7 emotions, 2 female speakers	Unseen cross-dataset test set

RAVDESS: <https://www.kaggle.com/datasets/uwrfkagglerravdess-emotional-speech-audio>

TESS: <https://www.kaggle.com/datasets/ejlok1/toronto-emotional-speech-set-tess>

3. Base Paper

J. H. Chowdhury, S. Ramanna, K. Kotecha, “Speech emotion recognition with light weight deep neural ensemble model using hand crafted features,” *Scientific Reports*, vol. 15, art. 11824, 2025. DOI: 10.1038/s41598-025-95734-z

Link: <https://www.nature.com/articles/s41598-025-95734-z>

4. Scope of the Project

In scope:

- **Preprocessing:** silence trimming, normalization, augmentation (noise, pitch shift, time stretch); mapping both datasets to a common emotion label set.
- **Feature extraction:** MFCC, Chroma STFT, Zero-Crossing Rate, RMS energy (librosa).

- **Models:** baselines (SVM, Random Forest) vs. 1D-CNN, CNN-BiLSTM and their ensemble (following the paper), using Scikit-Learn and TensorFlow/PyTorch.
- **Experiments:** (i) within-dataset evaluation on RAVDESS; (ii) cross-dataset evaluation (train RAVDESS → test TESS); (iii) feature ablation, removing one feature group at a time.
- **Evaluation:** accuracy, macro-F1, per-class confusion matrices, model size and inference time.
- **Bias check:** male vs. female accuracy and F1 on RAVDESS.
- **Deliverables:** reproducible codebase and a LaTeX technical report with results and error analysis.

Out of scope: large pretrained speech models (e.g., wav2vec 2.0, LLMs), real-time deployment, non-English or conversational (non-acted) speech. A simple Streamlit demo may be added only as a bonus.